# Week 5 — Archiving and Logging Data

## Overview

This week focused on two foundational sysadmin and security skills: **data archiving** for backup and recovery, and **log management** for audit trails and incident detection. Understanding how logs are generated, stored, and monitored is critical for any SOC or security operations role.

## Topics Covered

- File archiving and compression (`tar`, `gzip`, `bzip2`)
- Backup strategies and scheduling
- Linux logging system (`syslog`, `journald`, `rsyslog`)
- Log file locations and what they contain
- Log monitoring and analysis
- Introduction to `auditd` for system auditing

## Assignment

Completed a series of tasks involving:

- Creating compressed archives of directories and verifying their integrity
- Scheduling automated backups with `cron`
- Analyzing system logs to identify suspicious events
- Configuring `rsyslog` to forward logs to a remote log server

- Using `auditd` to track file access and privilege escalation events
- Writing a log analysis report documenting findings

## Key Commands & Concepts

### Archiving with tar

```bash
# Create a compressed archive
tar -czvf archive.tar.gz /path/to/directory

# Extract an archive
tar -xzvf archive.tar.gz

# List contents without extracting
tar -tzvf archive.tar.gz
```

### Important Log Files in Linux

| Log File | Purpose |
|---|---|
| `/var/log/auth.log` | Authentication events, sudo usage, SSH logins |
| `/var/log/syslog` | General system messages |
| `/var/log/kern.log` | Kernel messages |
| `/var/log/apache2/access.log` | Web server access requests |
| `/var/log/fail2ban.log` | Blocked IPs from brute force |

### Log Analysis Commands

```bash
# Search for failed SSH login attempts
grep "Failed password" /var/log/auth.log

# Count occurrences
grep "Failed password" /var/log/auth.log | wc -l

# Show last 50 lines of a log
tail -50 /var/log/syslog

# Follow a log in real time
tail -f /var/log/auth.log
```

### auditd — System Auditing

```bash
# Install and start auditd
sudo apt install auditd
sudo systemctl start auditd

# Watch a specific file for access
sudo auditctl -w /etc/passwd -p rwxa -k passwd_watch

# Review audit logs
sudo ausearch -k passwd_watch
```

## Security Relevance

Log management is the backbone of threat detection. Key security use cases:

- Detecting brute force attacks via repeated failed logins
- Identifying unauthorized file access
- Tracking privilege escalation (sudo events)
- Supporting post-incident forensic timelines
- Meeting compliance requirements (PCI-DSS, HIPAA, SOC 2 all require log retention)